BioInterpret: Evidence-Linked Interpretation of Genomic Models

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Abstract

We introduce BioInterpret, a framework that turns questions about genomic predictors into controlled computational experiments and evidence-linked interpretation reports. A configurable language model plans analyses, while a partition-aware executor fits task heads, freezes intervention plans, measures effects, and checks selected report claims against retained evidence. We evaluate 72 primary investigations across three DNA tasks, two pretrained encoders, four planning policies, and three repeats, with effect replay on separately reserved sequences. The structured LLM completes 17 of 18 reports; generic, fixed-ranking, and random-selection policies complete all 18 each. All 82 initially supported endpoints recur under the audit, but fixed ranking yields more replayed effects per investigation than the structured LLM (1.39 versus 1.22), at lower realized resource use. Enhancer results primarily reflect GC-composition dependence. A subsequent GenomeOcean-4B extension completes 36 further investigations, with all 33 supported endpoints replayed, but reveals numerical batch sensitivity and no clear planning advantage. A detailed promoter case and 18 synthetic runs further separate predictive accuracy, candidate recovery, and intervention response. The results establish an executable, inspectable interpretation workflow while showing that LLM planning does not automatically improve experiment selection. The measured effects describe fitted models under specified edits, not validated biological mechanisms.

1. Introduction

We introduce BioInterpret, a framework for investigating which sequence features a genomic predictor uses and reporting the evidence behind that interpretation. Genomic foundation models provide useful representations for regulatory-sequence tasks (Ji et al., 2021; Zhou et al., 2024). However, predictive accuracy alone does not tell a researcher whether a model responds to a regulatory motif, nucleotide composition, or another correlate of the training labels. These explanations can suggest different follow-up experiments even when they yield similar classification performance.

Practical interpretation still requires substantial coordination. A researcher must load the appropriate model, adapt its output to a task, select candidate features, construct valid controls, and decide whether an observed change supports the proposed explanation. Libraries such as GenomeFactory reduce the engineering burden of genomic-model development and expose representation-analysis tools (Wu et al., 2026). Attribution methods identify features associated with individual predictions (Lundberg & Lee, 2017; Sundararajan et al., 2017). Neither capability, by itself, specifies how to compare competing explanations under controlled sequence edits.

BioInterpret organizes this process around a simple interaction: select a genomic model, upload a labeled sequence dataset, and state the task and interpretation question in natural language. A configurable large language model (LLM) selects operations from a scientific tool registry. The executor performs the measurements and enforces data-access boundaries. Before evaluating a set of hypotheses, it freezes the candidate motifs, controls, sample budget, effect margin, and multiplicity family. The resulting report links each finding to an effect estimate, uncertainty interval, and raw intervention records.

The separation between reasoning and measurement is central to the design. The LLM can propose a plausible interpretation, but it cannot replace a numerical result or promote an inconclusive endpoint to a supported conclusion. Conversely, a valid model intervention need not reveal a biological mechanism: it measures the response of a particular predictor to a specified edit distribution. This distinction matters especially when a model has learned a spurious dependency or when the edits fall outside the biological training distribution.

The framework has three contributions. First, it pro-

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vides an implemented question-to-report workflow with interchangeable LLM endpoints and a common prediction interface, validated with DNABERT-2, original six-mer DNABERT, and GenomeOcean-4B. Second, it combines matched interventions, a frozen confirmation plan, and evidence-linked reporting in one executor. Third, it evaluates four planning policies in 72 investigations across three genomic tasks and two encoders, followed by a separate-sequence effect audit. A subsequent GenomeOcean extension adds 36 investigations with a causal decoder. An additional development case and 18 synthetic runs distinguish predictive performance, candidate recovery, and intervention stability.

The evaluation tests the planner rather than assuming its benefit. Structured and generic LLM policies share the executor and access checks with fixed-ranking and random-selection policies. All receive the same predictor-request caps; realized resource use is reported separately. An effect that recurs on reserved sequences supports a property of the fitted predictor under the specified edits. It does not establish a regulatory mechanism, and correlated endpoints do not constitute independent biological discoveries. The synthetic interpreter is deterministic, while sensitivity analyses of the development case are explicitly exploratory.

2. The BioInterpret Framework

2.1. User Interface and Model Access

The browser interface collects a registered genomic model, a dataset ZIP, and a free-text task description. Users do not provide server filesystem paths. A ZIP contains one UTF-8 CSV with sequence and binary-label columns; optional sample identifiers and explicit split assignments support provenance. Upload validation rejects unsafe archive paths, duplicate or reverse-complement sequences, invalid bases, and partitions without both classes. The current input contract accepts DNA sequences of 12–1,000 bases, subject to the encoder's stricter token limit.

Model configuration belongs to the host. Three validated adapters load DNABERT-2, original six-mer DNABERT, and the GenomeOcean-4B causal decoder and fit logistic classification heads. An additional adapter accepts standard Hugging Face DNA binary classifiers with existing trained heads; it rejects missing or mismatched parameters. The interface lists host-configured models rather than claiming that every genomic architecture is interchangeable. Completed reports include follow-up chat grounded in the saved evidence and conversation history. This mode answers questions without executing new tools; a user can carry the latest question into a new investigation when new measurements are needed. LLM provider, endpoint, and model identifiers are configurable independently of the genomic predictor. The measured LLM investigations use GLM-5.3 through a compatible Chat Completions endpoint.

2.2. Task Adaptation and Data Boundaries

An investigation uses training, validation, exploration, and confirmation partitions. All three genomic backbones are frozen. We pool its final token states using the number of covered bases as weights, excluding special tokens. Original DNABERT instead receives overlapping six-mers and uses mean final-token pooling, also excluding special tokens. GenomeOcean uses base-length-weighted final causal token states, excluding padding and special tokens. The task head standardizes training embeddings and fits logistic regression with regularization $C \in \{0.01, 0.1, 1\}$; validation log loss selects C. Predictions use the probability of label 1. The tokenizer must cover the full sequence. Inputs exceeding the supported token context (512 for the BERT encoders; 10,240 for GenomeOcean) are rejected rather than silently truncated.

Exploration tools operate before confirmation is opened. They rank all six-mers by association with model probability after residualizing both counts and predictions on GC content and sequence length. Candidates must occur in at least five exploration sequences. Additional tools scramble local windows and compute single-base substitution effects. These measurements generate hypotheses; their candidate ranking and enrichment statistics are not confirmatory tests. Two cached JASPAR motif matrices, TBP and SP1, supply targeted promoter priors. Consensus similarity is not a significant PWM match or an assignment of transcription-factor identity.

2.3. LLM Planning and Bounded Execution

The planner calls nine typed tools: dataset profiling, model preparation, motif discovery, window occlusion, reference lookup, experiment design, experiment execution, saturation mutagenesis, and report writing. It receives structured outputs and may revise arguments after an error. Tools do not execute arbitrary LLM-generated code. The run is capped at 14 LLM turns, with bounded per-tool inputs and at most three motif hypotheses plus a composition contrast. This is a tool-using policy prompted to consider alternatives; the implementation does not optimize an explicit information-gain objective.

A valid experiment plan records the rationale, competing hypotheses, motifs, matched controls, sample budget, minimum effect size, and family-wise error target. The executor hashes the plan before opening confirmation. It permits one confirmation call and prevents renewed discovery or plan changes afterward. The LLM sees the resulting evidence to write the report, but cannot use it to select another confirmation experiment within the same study. This

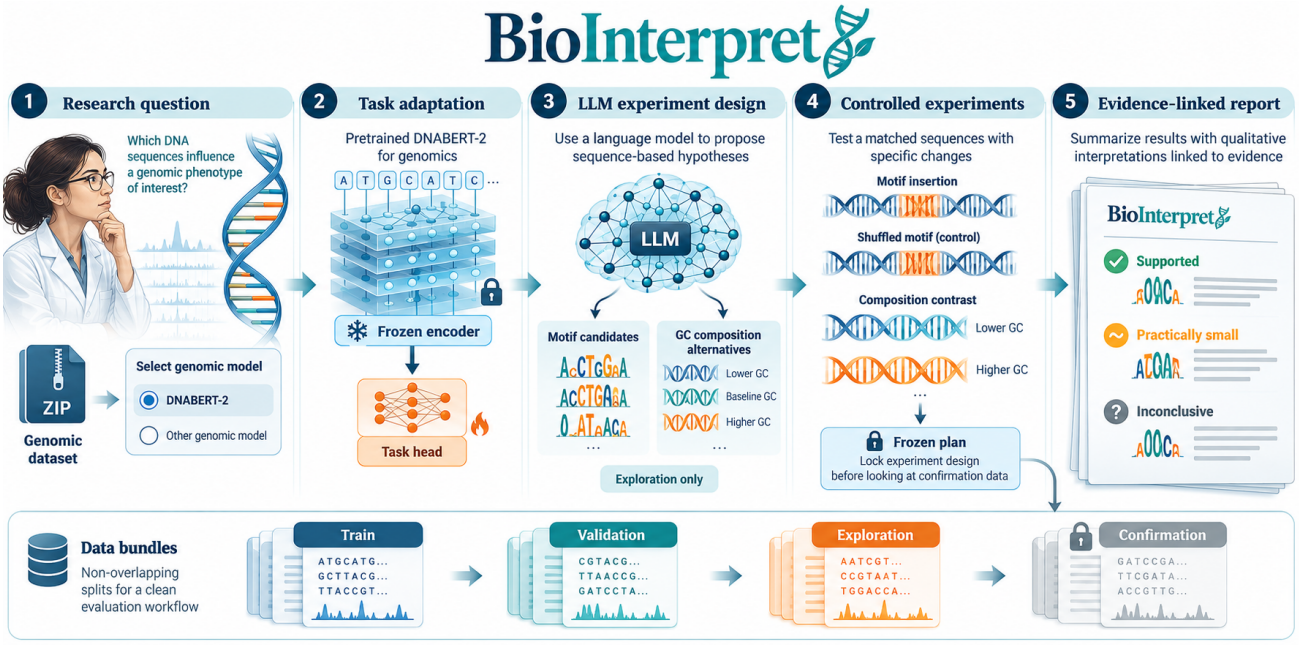


Figure 1. The implemented BioInterpret workflow. A user supplies a research question, a model choice, and labeled sequences. The executor fits a task head where needed, exposes exploration tools to an LLM, freezes the intervention plan, and evaluates it on a separate confirmation partition. Reports link conclusions to measured evidence. The four data bundles are disjoint partitions, not successive transformations of the same examples. This GPT-generated conceptual illustration contains no quantitative results; sequence glyphs and neural layers are schematic.

is a software-enforced within-episode boundary, not an external preregistration or protection against repeated analyst use of a dataset.

2.4. Interventions and Decision Rules

For predictor probability s_f, intervention T, control C, background population P, and paired edit randomness ω, the target contrast is

$$\Delta_f(T, C; P) = \mathbb{E}_{x \sim P, \omega}[s_f(T(x; \omega)) - s_f(C(x; \omega))]. \quad (1)$$

For a motif, the executor inserts the candidate and a distinct shuffled version at the same position in the same source sequence. The pair has identical base counts. Source sequences containing the motif are excluded; accepted treated sequences contain exactly one occurrence and controls contain none. The comparison isolates sequence order relative to that shuffle distribution, while still potentially changing other motifs and local structure.

The composition contrast changes the background to approximately 65% versus 35% GC while preserving the first selected candidate's sites and occurrence count. It does not preserve all sequence features, and therefore cannot identify a uniquely causal biological role for GC. Both experiments use mixed-label backgrounds. An intervention effect is a property of the predictor under these edits, not evidence that the same edit changes transcription in cells.

We compute paired bootstrap intervals using 2,000 resamples in the original study. For m frozen endpoints, each interval has nominal level $1-0.05/m$. With a practical-effect margin $\delta = 0.03$, an interval wholly above δ or below $-\delta$ supports a model effect; an interval wholly inside $[-\delta, \delta]$ supports a practically small effect. All other intervals are inconclusive. Bonferroni allocation addresses the within-study endpoint family, conditional on valid marginal intervals; it does not make finite-sample percentile bootstrap coverage exact or adjust repeated development studies.

2.5. Evidence Ledger and Reporting

Every accepted pair records its source identifier, edited sequences, probabilities, effect, plan hash, and split hash. The manifest records model adaptation, configuration, and source hashes. Failed calls remain in the event trace. Numeric tables are assembled directly from evidence records. The report tool requires existing evidence identifiers and exact agreement with stored endpoint decisions; it also checks selected population and uncertainty wording. These rules caught errors in the measured runs, but do not provide exhaustive semantic verification. A researcher must still review the report, including the hypothesis rationale and biological interpretation.

3. Experimental Evaluation

3.1. Questions and Evaluation Design

We evaluate three questions: can the system complete an interpretation investigation across genomic tasks and pretrained encoders; do selected effects recur on a separate sequence population; and does the structured planner improve on simpler policies under the same execution caps? The primary experiment crosses three tasks, two encoders, four policies, and three repeats, yielding 72 assigned investigations. A subsequent causal-decoder extension adds 36 investigations. A separate development case supplies detailed intervention-sensitivity analyses, and 18 synthetic model runs test the measurement machinery.

Tasks and fixed predictors. The tasks are human promoter classification from the DNABERT authors' example, human enhancer classification from Cohn, and fruit-fly enhancer classification from Stark through Genomic Benchmarks (Ji et al., 2021; Grešová et al., 2023). The two additional datasets are distinct from the earlier promoter development case. Each task retains its supplied train/test boundary. From training data we select 1,600 training, 320 validation, and 256 exploration sequences; from test data we reserve 256 confirmation and 512 audit sequences. Sequences longer than 500 bases are explicitly center-cropped before exact and reverse-complement deduplication. This affects most fruit-fly sequences, so our results are for the cropped task rather than the full-context benchmark.

We fit one task head per dataset–encoder cell using the same validation rule for DNABERT-2 and original six-mer DNABERT. All policies and repeats within a cell share the fitted predictor and exact partition membership. The six cells test reuse across two BERT-family encoders and three related DNA tasks, not broad modality transfer. Dataset identities, checkpoint revisions, prompts, split hashes, and the evaluation protocol are retained. The original promoter data were used during development; the external enhancer outcomes were not used to tune this evaluation protocol.

Policies and matched caps. *Structured LLM* uses BioInterpret's workflow and competing-hypothesis prompt. *Generic LLM* uses a neutral biomedical-assistant prompt with the same LLM, scientific caution, reporting requirements, tool schemas, and access rules, but without the prescribed workflow. *Fixed ranking* selects the three highest-ranked exploration six-mers. *Random selection* samples three of the same top-20 candidates. Both deterministic policies include the GC contrast and write templated reports through the same checks. Consequently, these comparisons evaluate planning policies within the executor; they do not compare the full system against an unconstrained agent with weaker safeguards.

All methods are capped at 8,000 requested predictor sequences per episode, counted before caching, and share per-tool quotas. The two LLM policies share a 14-turn cap and 4,096 generated tokens per call. These are matched caps, not matched realized token or compute consumption. We record input and output tokens, predictor requests, failures, and report retries. Predictor fitting is shared preparation outside the planner budget. Policy order is rotated, but shared caches and provider latency prevent interpreting wall time as a controlled accelerator-efficiency comparison.

Separate effect audit. After all episodes in a cell have ended, an evaluator opens its 512 reserved sequences and replays each selected contrast with two independent edit seeds. The predictor and intervention definitions remain fixed; results are never returned to the planner. Effects are averaged within source across edit seeds, then source means are bootstrapped using the original endpoint-family interval level. A supported effect is *replayed* if the audit interval also lies outside the practical-null band in the same direction. We report this rate together with supported-effect count, practical-decision agreement, and replayed effects per assigned episode. Completed measurements remain in the audit even if report generation fails; report completion is a separate endpoint. Repeated endpoints are correlated, and a replay is not biological ground truth.

3.2. Completion, Effect Replay, and Resources

The structured policy completed 17 of 18 reports; the other policies completed all 18 (Table 1). One structured promoter/DNABERT-6 episode failed after measurements because the provider response was truncated or filtered. It remains in the assigned denominator and its completed endpoints remain in the audit. The six predictors' confirmation AUROCs range from 0.638 to 0.868 (Table 2); interpretation yield should not be read as predictive superiority.

Across policies, all 82 initially supported endpoints were supported in the same direction on audit sequences. The structured, generic, fixed, and random policies contributed 22, 19, 25, and 16 of these endpoints, respectively. Their replayed yields were 1.22, 1.06, 1.39, and 0.89 per assigned episode. These are repeated endpoint counts, not 82 independent replications: policies often select the same contrast, and every cell shares a predictor. No superiority claim or independent-binomial confidence interval is warranted from these pooled counts.

Task dependence limits the planning result. All 34 supported promoter endpoints were motif-order effects, whereas all 48 supported enhancer endpoints were GC contrasts. Every policy obtained exactly one replayed effect per episode on each enhancer cell. The structured planner therefore did not improve yield on those tasks. Differ-

Policy	Complete	Supported	Replayed	Yield	Tokens	Pred. seq.
Structured LLM	17/18	22	22	1.22	41,388	2,346
Generic LLM	18/18	19	19	1.06	32,944	2,406
Fixed ranking	18/18	25	25	1.39	0	1,418
Random selection	18/18	16	16	0.89	0	1,454

Table 1. Paired policy evaluation. Supported and replayed are endpoint counts, not independent biological discoveries. Yield is replayed supported endpoints per assigned episode, including failures. Tokens and predictor sequences are medians per episode; input and output tokens are included. Deterministic policies use no LLM. All methods share hard access and reporting checks.

Task	DNABERT-2	DNABERT-6
Promoter	0.795	0.868
Human enhancer	0.791	0.769
Fly enhancer	0.638	0.705

Table 2. Confirmation AUROC for the six fixed predictors in the paired evaluation. Each value uses 256 sequences; all policies inspect the same fitted predictor within a task–encoder cell.

ences were concentrated in promoter selection: fixed ranking matched the structured policy with DNABERT-2 and exceeded it with original DNABERT (Figure 3). Selection stability also varied: structured motif-set Jaccard ranged from 0.30 to 1.00 across cells, while fixed ranking was identical across repeats by construction. Stability alone does not establish scientific utility.

The audit resolves uncertainty without changing supported conclusions. Of 287 measured endpoints, 143 were practically small in both populations, 82 were supported in both, and 38 remained inconclusive. Another 12 changed from inconclusive to supported and 12 from inconclusive to practically small. Thus 263/287 decisions agreed. The larger audit population and averaging over edit seeds differ from the original confirmation design; resolving uncertainty is not the same as disproving the earlier inconclusive decision. Endpoint-specific intervals and raw records are retained.

Reporting and resource costs remain material. The structured policy had 26 rejected report attempts across 12 episodes, compared with 20 across 11 generic episodes. Of the 46 rejections, 26 concerned inconclusive wording or overstatement, 12 attempted numerical prose, seven disagreed with a stored verdict, and one had malformed arguments. These trace-level observations identify checks that fired, not the checks' sensitivity to all scientific errors. Median total LLM tokens were 41,388 for structured and 32,944 for generic; deterministic policies used none. Median requested predictor sequences were 2,346, 2,407, 1,418, and 1,454, respectively. The fixed policy's higher pooled yield at lower realized cost argues against assuming that a more elaborate planner is necessary for this candidate-selection benchmark. The incomplete provider response's token usage was unavailable, so failed-call token accounting is incomplete.

3.3. GenomeOcean-4B Architecture Extension

After inspecting the primary evaluation, we added the original GenomeOcean-4B checkpoint (Zhou et al., 2025). This microbial-pretrained causal decoder extends the implementation beyond BERT encoders. We froze a separate extension protocol before running 36 additional investigations: the same three tasks, four policies, and three repeats. The original 72 episodes and their protocol remain unchanged. This is an exploratory architecture extension on already inspected tasks, not a prospectively frozen three-model benchmark.

The frozen decoder uses base-length-weighted final causal states and the same logistic-head selection rule. Splits, edit rules, prompts, and prediction caps match the primary evaluation, including the explicit 500-base crop. These short eukaryotic tasks test cross-domain adaptation rather than microbial long-context generation. The audit again holds the newly fitted predictor fixed.

All 36 reports completed, and all 33 initially supported endpoints replayed in the same direction (Appendix F). Structured, generic, fixed, and random policies yielded 10, 9, 9, and 5 replayed endpoints: 1.11, 1.00, 1.00, and 0.56 per assigned episode. The structured policy's one-endpoint advantage over fixed ranking is descriptive, not evidence of general superiority. It also incurred 16 report rejections and a median 44,213 total LLM tokens; generic incurred 14 rejections and 41,460 tokens. Practical decisions agreed for 114/144 endpoints; 30 initially inconclusive endpoints resolved to supported (22) or practically small (eight) in the audit.

GenomeOcean confirmation AUROC was 0.840 on promoters, 0.706 on human enhancers, and 0.626 on fly enhancers. Its promoter differences from both BERT encoders had exploratory paired 95% intervals crossing zero. On human enhancers, its difference from DNABERT-2 was -0.085 $[-0.136, -0.035]$. Probability-error metrics and full paired intervals are retained. These results show that the executor transfers to a larger causal backbone, while neither scale nor successful report completion guarantees better predic-

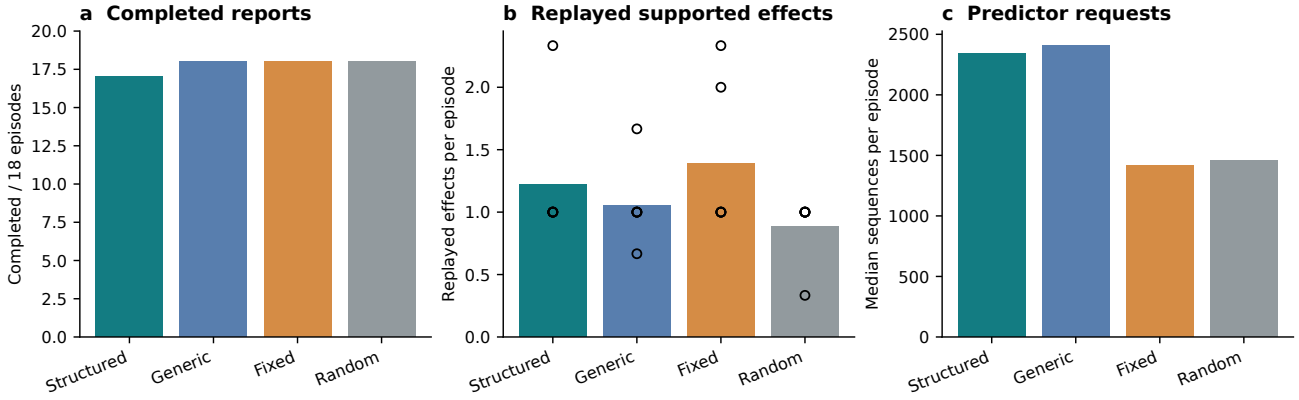


Figure 2. Measured policy outcomes across the 72 assigned investigations. Each policy has 18 episodes. Replayed supported effects require a compatible practical effect on reserved audit sequences. Open circles in the middle panel are the six task–encoder cell means, each averaging three repeats; they are not confidence intervals. Predictor requests count sequences before cache lookup. LLM and deterministic policies have the same access checks and prediction caps, while their realized computation differs.

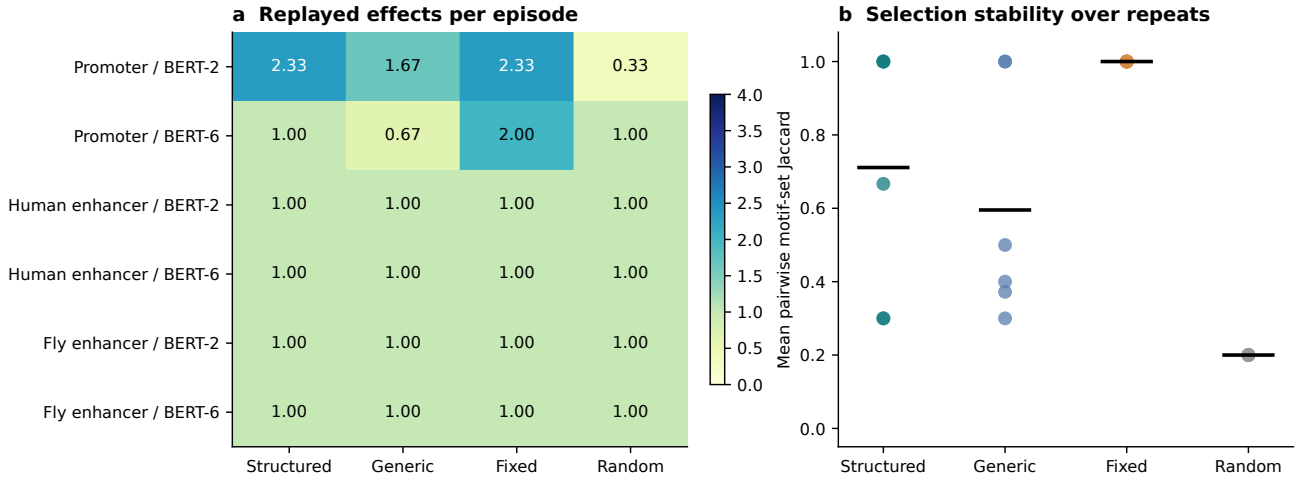


Figure 3. Task dependence and repeat stability. (a) Replayed supported effects per assigned episode, averaged across three repeats in each task–encoder cell. This measures audit-consistent model effects, not biological discoveries. (b) Pairwise Jaccard overlap of selected motif sets across repeats; each dot is one cell, and the black segment is the six-cell mean. Stable selection need not imply useful or biologically meaningful effects.

tive or interpretation performance.

Mixed-precision numerical sensitivity. Replaying 36 sampled predictions with reconstructed original inference batches reproduced stored probabilities within 2.3×10^{-16}. Rebatching those inputs, however, produced task-wise maximum differences of 0.00201, 0.00938, and 0.00793, exceeding the prespecified 0.002 tolerance in all three settings. We retain these failed diagnostics. A post-hoc FP32 replay of all 37,270 audit pairs preserved all 144 endpoint decisions with fixed heads and edits; sampled FP32 batch differences were below 1.6×10^{-6} (Appendix F.1). Individual BF16–FP32 probability shifts reached 0.111, so stable audit decisions do not imply invariant predictions.

3.4. Detailed Development Case

The earlier promoter episode used a different retained head and split configuration: 2,000 training, 400 validation, 320 exploration, and 320 confirmation sequences. Its AUROC was 0.842, and its four frozen endpoints produced one supported, one practically small, and two inconclusive effects. CTAGTG insertion had a mean effect of 0.0797 with a 98.75% interval [0.0485, 0.1121] against a composition-matched shuffle. The GC contrast was inconclusive at the 0.03 practical threshold. This development case is distinct from the 72-episode evaluation and is not pooled into its primary results.

Follow-up analyses sharpened the interpretation. A six-mer logistic baseline outperformed that frozen DNABERT-2 head on the inspected split (AUROC 0.891 versus 0.842).

The aggregate GC effect concealed opposite subgroup means: -0.0754 on non-promoters and $+0.2393$ on promoters. Across 12 CTAGTG edit conditions, means remained positive at 0.0604–0.1063, whereas refitting the task head changed some endpoint decisions. These analyses show why predictor choice, background population, comparison distribution, and head fitting belong in the evidence record. They are exploratory, reuse inspected data, and do not independently validate a regulatory mechanism (Appendix D).

3.5. Synthetic Controls and Failure Interpretation

The synthetic benchmark has three regimes (motif, composition-confounded, and label-independent), two predictor families (CNN and random forest), and three seeds. A deterministic interpreter recovers the planted candidate in all 12 non-null runs. All six confounded predictors show supported composition effects and lose discrimination under a shift that removes the GC–label association. However, near-chance null forests still have measurable responses to selected motif insertions, and one accurately predicting CNN has a practically small insertion contrast for the recovered motif. Thus candidate recovery, task accuracy, and predictor intervention response are distinct quantities. These 18 runs validate controls and expose failure modes; they do not test LLM planning quality (Appendix C).

What the checks establish. Numerical verification reconstructs intervention differences, tests composition and occurrence constraints, checks partition and plan hashes, and replays model predictions. Report checks reject selected decision, population, and numerical inconsistencies. They do not assess all biological assertions, and the application still permits scientifically imprecise free-text planning rationales. This is a within-trace mechanism audit, not a blinded estimate of semantic error detection or a randomized ablation of the reporting rules. All policies retain the same hard checks so that planning differences are not confounded with different safety or data-access privileges.

4. Related Work

Genomic models and interpretation libraries. DNABERT learns DNA representations and demonstrates regulatory-sequence interpretation (Ji et al., 2021). DNABERT-2 introduces a more efficient sequence representation and a multi-species evaluation suite (Zhou et al., 2024). GenomeOcean uses causal modeling of assembled metagenomes (Zhou et al., 2025). Genome-Factory integrates data preparation, tuning, inference, benchmarking, and sparse-autoencoder-based interpretation across genomic models (Wu et al., 2026). BioInterpret addresses a complementary interaction: a user asks an interpretation question, and an LLM coordinates controlled predictor experiments. It currently has substantially narrower validated model coverage than Genome-Factory.

Attribution and model dependence. SHAP and Integrated Gradients formalize feature attribution (Lundberg & Lee, 2017; Sundararajan et al., 2017). These methods can generate useful candidate features, but attribution scores and matched intervention contrasts answer different questions. BioInterpret does not introduce a new attribution estimator. It makes the intervention, comparison distribution, and evidential status explicit in a question-to-report workflow.

Biomedical agents. Biomni integrates biomedical tools with LLM reasoning (Huang et al., 2025), and BioDiscoveryAgent uses an agent to select genetic perturbation experiments (Roohani et al., 2025). Tool use, iterative reasoning, and experimental planning are therefore established ideas. Our system focuses on interpreting an existing predictor through bounded interventions and enforcing consistency between measured endpoint decisions and report fields. Our generic-policy comparator shares the same model, tools, and resource caps; it is not an evaluation against the full Biomni or BioDiscoveryAgent systems.

5. Discussion and Conclusion

BioInterpret turns a genomic-model question into executed interventions and an evidence-linked report. The evaluation spans three DNA tasks, two pretrained encoders, four planning policies, and three repeats per setting. A subsequent 36-episode extension evaluates GenomeOcean-4B. Separate audit sequences test whether selected model effects recur beyond the agent's confirmation backgrounds. The detailed promoter case and synthetic controls show why task accuracy, candidate recovery, and intervention response need separate evaluation.

The comparison does not establish a general advantage for LLM planning. A fixed ranking policy can recover audit-consistent effects at lower realized cost, and report generation remains a source of retries and failure. The practical contribution is an inspectable research workflow: researchers can state a question, compare controls, inspect uncertainty, and continue the discussion without silently revising the completed experiment. The policy comparison identifies where stronger planning objectives and more discriminating tasks are needed.

The evaluation remains limited to related binary DNA tasks and frozen representations from two BERT encoders and one causal decoder. The GenomeOcean extension was designed after inspecting the primary results and does not test microbial long-context generation. Random sequence

partitions and exact deduplication do not address genomic proximity, homology, or pretraining contamination. Most fly sequences are cropped to the common context length. GenomeOcean exhibits bfloat16 batch sensitivity; a fixed-head FP32 follow-up preserves all audit decisions but does not establish invariance of individual predictions. The audit holds the trained predictor fixed, so it does not measure robustness across independent model training or establish biological causality. Motif insertion and broad GC edits can move inputs outside the biological distribution. Repeated endpoints and shared datasets limit independent statistical replication.

The reporting checks enforce selected consistency rules rather than complete semantic correctness. A stronger study should add coordinate-grouped datasets, additional architecture families, blinded report assessment, and independently defined experiment utility. Claims of regulatory discovery require biological validation beyond this computational audit. The current evidence supports executable, traceable model investigations and a concrete comparison with simpler policies; it does not justify treating plausible agent prose as scientific evidence.

Impact Statement

This work aims to make computational interpretation of genomic predictors easier to execute and inspect. Its main risk is that a coherent automated report could be mistaken for experimentally validated biology. We therefore distinguish model responses from biological effects and retain uncertainty and failed attempts. The current experiments use public or synthetic DNA data and do not evaluate clinical decisions. Deployments accepting private genomic datasets require appropriate access controls and data governance; the prototype is not an authenticated multi-user service. GPT Image was used for the conceptual workflow illustration. All quantitative figures are generated from retained numerical outputs.

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A. Data, Models, and Reproduction

The promoter data are the original DNABERT repository's six-mer sample data at revision `b6da04e`. The DNABERT-2 weights use revision `7bce263`. The release includes acquisition code, the source manifest, and retained study artifacts. The data's train/development boundary is preserved before stratified subsampling. Exact and reverse-complement duplicates are removed globally. No chromosome, locus, or homology grouping is available in the current sample contract; this limits the interpretation of held-out performance.

The reference implementation uses Python 3.8, PyTorch 2.4, Transformers 4.28, NumPy 1.24, and scikit-learn 1.3. DNABERT-2 inference uses native PyTorch attention, with the legacy Triton path disabled. The trained logistic head stores feature means, scales, coefficients, and intercept. The same task-head representation is used when replaying predictions. The case's GLM endpoint is `z-ai/glm-5.3`; API calls and reported provider identifiers are retained, but a hosted model alias does not guarantee an immutable provider-side checkpoint.

The canonical CLI entry point for an investigation is `python -m bointerpret.agent`. The scientific audit is `scripts/verify_study.py`. Additional analyses are generated by `scripts/analyze_paper.py`, and the development-case quantitative figures and predictor table are rendered by `scripts/render_paper_analysis.py`. The additional analyses require the saved study, pretrained weights, and a scientific Python environment. The original 18 synthetic runs use `scripts/run_benchmark.sh`. The web interface and CLI share the same executor.

B. Detailed Exploratory Analysis Protocol

The follow-up design is saved before execution, but after the original results were inspected. It is not an independent preregistration. Predictor AUROC intervals use 4,000 stratified bootstrap resamples of the 320 confirmation rows. The same sampled indices yield paired AUROC differences. Brier score, log loss, and accuracy use the full retained confirmation partition; accuracy thresholds probability at 0.5. The six-mer baseline includes GC fraction in addition to the 4,096 counts, and all non-null logistic baselines select C from the same three values by validation log loss. The negative-control head shuffles training labels once with seed 42 and fixes $C = 0.01$.

Subgroup and edit-sensitivity intervals use 4,000 ordinary paired bootstrap resamples and 95% marginal coverage, without correction across the post-hoc analyses. Head-refit intervals use 98.75% marginal coverage for comparability with the original four endpoints, but do not correct across refits or analyses. The three training bootstraps sample training rows with replacement using seeds 7, 17, and 29. They retain the original validation partition and select regularization again. These repeated fits share both data and encoder and are not independent experimental replicates.

For CTAGTG position sensitivity, left insertion begins at zero-based position 8, center insertion begins at $\lfloor (L - 6)/2 \rfloor$, and right insertion begins at $L - 14$. Random insertion samples uniformly from valid start positions. Shuffled controls use the same position and exact base counts. All accepted treated sequences have exactly one CTAGTG occurrence, and controls have none. The candidate-free source population is unchanged across conditions. Three seeds crossed with four policies produce 1,428 pairs. Additional nested-prefix analyses at 32, 64, and all available pairs are retained in the machine-readable results as descriptive sample-size checks; no stopping rule or power claim is derived from them.

C. Synthetic Benchmark Details

Each seed and regime has 3,000 training, 600 validation, 600 exploration, 800 IID test, and 800 shifted examples. Motif occurrence probabilities in non-null tasks are 0.95 and 0.05 for the two classes. In the confounded regime, high versus low GC tracks the class with probability 0.97; the shifted set removes this dependence. The null regime uses motif occurrence probability 0.5 independently of class and label-independent GC. Generated sequences are rejected and redrawn when their realized motif counts do not match the construction.

The CNN has 32 width-six convolutional filters, max and mean pooling, a 32-unit hidden layer, and a binary output. It trains for up to 16 epochs with validation checkpoint selection. The random forest has 100 trees. The deterministic interpreter screens six-mers on exploration, evaluates eight candidates with validation interventions on up to 160 sources, and freezes two endpoints before opening test data. Each endpoint uses up to 400 sources, with 2,000 paired bootstrap resamples and 97.5% marginal intervals. The practical-effect margin is 0.05, distinct from the promoter case's 0.03. Seeds are 7, 17, and 29; the shared datasets pair model families within a regime and seed.

D. Development Case and Sensitivity Analyses

D.1. Evaluation Design

We organize evaluation into three parts: a real-data question-to-report episode, controlled tests of the intervention machinery, and sensitivity analyses of the real-data conclusions. These evaluate execution and model behav-

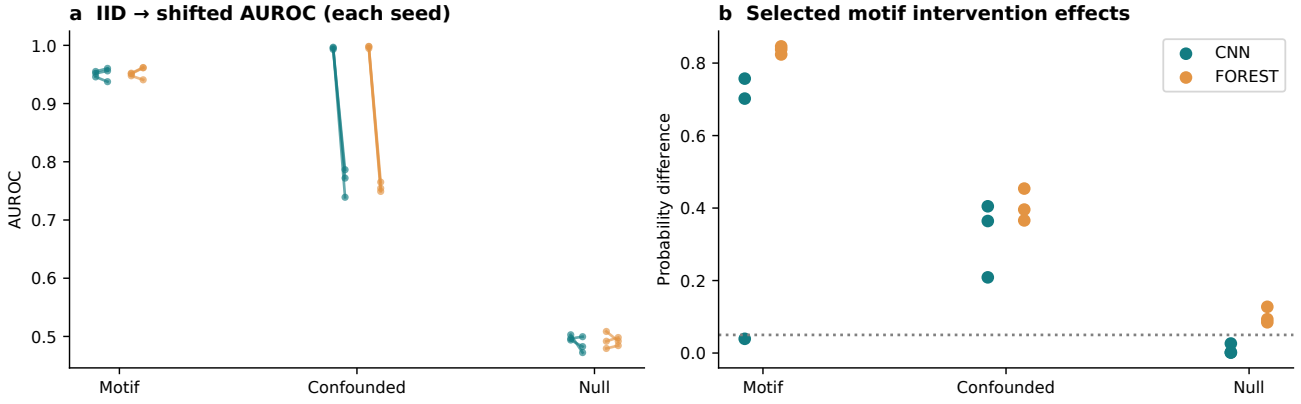


Figure 4. Synthetic controls for the deterministic interpretation pipeline. (a) Each short segment connects IID to shifted AUROC for one trained model; horizontal separation only indicates direction. Confounded models lose discrimination under the GC shift. (b) Selected motif intervention effects vary across families and regimes, including nonzero effects in near-chance null forests. Dots are individual runs; the dotted line is the synthetic practical margin of 0.05.

ior. They do not constitute a matched comparison of LLM planning policies.

The real case uses the DNABERT authors' public promoter example (Ji et al., 2021). Reconstructing overlapping six-mers and removing 25 exact or reverse-complement duplicates leaves 33,341 labeled sequences. We preserve the supplied train/development boundary: 2,000 training and 400 validation examples come from training data, while 320 exploration and 320 confirmation examples come from development data. The subsets are stratified with seed 42. They are disjoint, but the source lacks a genomic-coordinate split audit. DNABERT-2 is frozen and receives a fitted task head; this adapts the original paper's interpretation question without reproducing its attention and motif-matching pipeline.

The completed GLM episode made ten LLM calls. It used the exploration outputs and promoter priors to select GGCGCT, CTAGTG, and TATAAA, together with a background-GC contrast. The four endpoints requested 128 backgrounds each; admissibility filtering yielded 497 matched pairs in total. Because this dataset was used in earlier development runs, even the original within-episode confirmation is an initial case study rather than a fresh external validation. All additional analyses below are explicitly exploratory and reuse the same split.

D.2. Predictive Performance and Negative Controls

The saved DNABERT-2 head achieves AUROC 0.842 with a stratified bootstrap 95% interval of [0.799, 0.884] and accuracy 0.759 on 320 sequences. To contextualize this performance, we fit logistic models using either GC fraction alone or the 4,096 six-mer counts plus GC fraction. Their regularization is selected by validation log loss using the same candidate grid as the foundation-model head. A negative-control head uses the DNABERT-2 embeddings, shuffled training labels, and fixed $C = 0.01$, without tuning against true validation labels.

Predictor	AUROC	Accuracy	Brier
DNABERT-2	0.842	0.759	0.162
GC logistic	0.478	0.494	0.250
6-mer logistic	0.891	0.812	0.137
Permuted-label head	0.507	0.519	0.287

Table 3. Exploratory predictor comparison on the same 320 confirmation sequences. All trained heads use the same training partition; validation selects regularization except for the fixed, permuted-label negative control.

The six-mer baseline achieves AUROC 0.891, exceeding the frozen foundation-model head on this split (Table 3). The paired bootstrap interval for DNABERT-2 minus six-mer AUROC is [-0.095, -0.003]. This descriptive comparison does not rank fully fine-tuned models or establish a general advantage of six-mers. It shows that a simple task-specific baseline is essential when interpreting the chosen head. The GC-only model and permuted-label head achieve AUROCs 0.478 and 0.507, respectively. Weak GC-only prediction does not imply that changing GC cannot affect a nonlinear sequence model.

D.3. The Original Interpretation Episode

CTAGTG insertion increases model probability relative to its base-matched shuffle by 0.0797, with interval [0.0485, 0.1121] across 119 pairs. The entire interval exceeds the 0.03 practical threshold, supporting a positive sequence-order effect for this predictor and edit distribution. This does not identify a transcription factor or demonstrate a regulatory effect in cells.

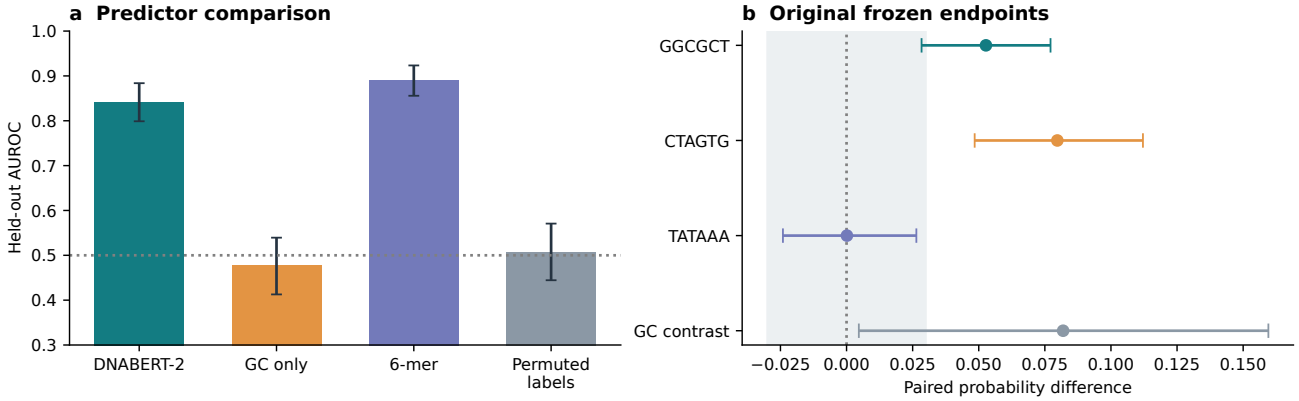


Figure 5. Promoter prediction and original intervention outcomes. (a) Predictor AUROC with exploratory 95% stratified bootstrap intervals from 4,000 resamples. (b) Original frozen endpoint effects with 98.75% paired bootstrap intervals, allocating nominal family-wise error 0.05 across four endpoints. The shaded region is the practical-null band $[-0.03, 0.03]$. The two panels use different interval levels and answer different questions.

GGCGCT has a mean effect of 0.0527 with interval [0.0284, 0.0771] across 124 pairs. Although the interval is positive, it crosses the practical threshold and is classified as inconclusive under the frozen rule. TATAAA has mean 0.0001 and interval [-0.0241, 0.0264] across 126 pairs, supporting a practically small average effect under random-position insertion. It does not show that TATA elements are biologically unimportant or that the predictor ignores them at every position. The GC contrast has mean 0.0819 and interval [0.0046, 0.1595] across 128 pairs, also inconclusive at the practical threshold. Figure 5 shows the four outcomes without turning positive point estimates into uniformly supported claims.

D.4. Backgrounds, Controls, and Intervention Stability

Background heterogeneity. Post-hoc stratification reveals why the aggregate GC contrast deserves caution (Figure 6a). Its mean is -0.0754 on non-promoter backgrounds and $+0.2393$ on promoter backgrounds. The respective descriptive 95% intervals are [-0.1444, -0.0032] and [0.1669, 0.3174], with 64 sequences in each stratum. CTAGTG remains positive in both groups, with means 0.0835 and 0.0754. These are reused-data subgroup analyses without multiplicity adjustment; they motivate an independently frozen heterogeneity test rather than establish a biological interaction.

Choice of control. We recompute contrasts against each unedited source, preserving the treated sequences from the original ledger (Figure 6b). CTAGTG's mean changes from 0.0797 against matched shuffles to 0.0947 against unedited backgrounds. For the GC experiment, the high-versus-low contrast is 0.0819, whereas high-GC versus unedited is 0.0196. These are different estimands, not competing estimates of an identical effect. In particular, the unedited comparison does not isolate motif order and can include the effects of composition and the displaced sequence.

Edit position and randomness. We repeat CTAGTG insertion using three edit seeds and four position policies: left, center, right, and uniformly sampled positions. Every policy uses the same first 128 confirmation backgrounds, excludes existing occurrences, and verifies exact base-count matching. Each of the 12 conditions retains 119 pairs. Mean differences range from 0.0604 to 0.1063; random-position means range from 0.0783 to 0.1019 (Figure 6c). The positive direction persists in these conditions, but magnitude depends on the edit design. The 1,428 new pairs are correlated repeated measurements on the same sources, not 1,428 independent biological samples.

Task-head stability. We bootstrap the 2,000 training rows three times, refit the head, and repeat validation-based regularization selection while keeping the encoder and confirmation backgrounds fixed. AUROCs range from 0.818 to 0.824. CTAGTG mean effects remain positive at 0.0639–0.0747, whereas GC effects vary from 0.0431 to 0.1472 (Figure 6d). The TATAAA intervals extend outside the practical-null band for these refits. Thus, the original practically-small decision is specific to the saved head; it is not a stable property of all task heads fitted to this dataset. These three bootstraps assess a narrow form of training-sample sensitivity, not between-dataset generalization.

D.5. Controlled Synthetic Tasks

The retained synthetic benchmark tests the intervention machinery with three regimes, two predictor families, and three seeds, producing 18 model runs. Sequences have 96 bases and a planted six-mer CACGTG. The motif regime associates labels with motif presence; the confounded

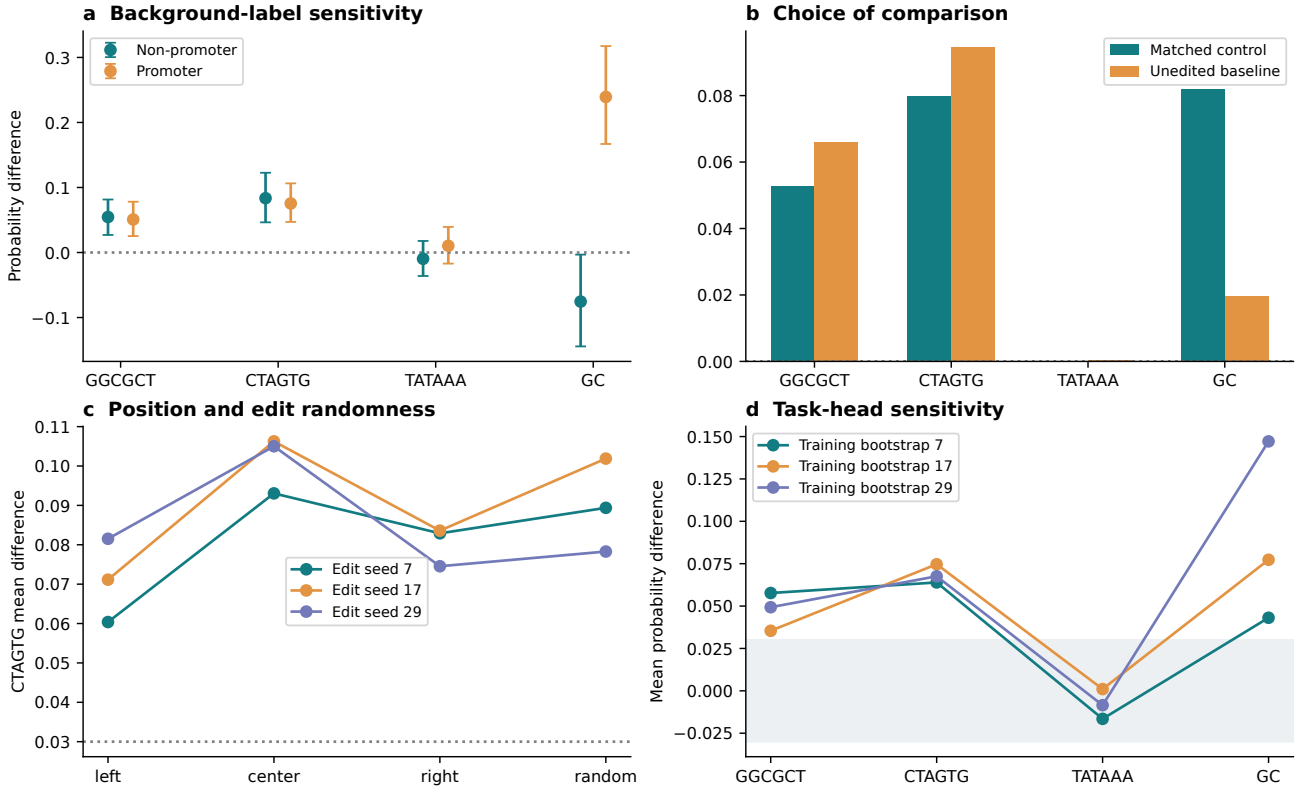


Figure 6. Exploratory sensitivity analyses on the inspected promoter split. (a) Background-label strata with unadjusted 95% paired bootstrap intervals. (b) Matched and unedited-source comparisons estimate different effects. (c) CTAGTG effects across positions and three edit seeds. (d) Re-fitting the task head on three training bootstraps changes effect magnitudes. Panels (b)–(d) show means; connected points and repeated seeds are not independent task replications.

regime additionally associates labels with GC content; the null regime makes both features label-independent. A shifted set removes the GC–label association. The predictors are a small sequence CNN and a random forest on six-mer counts and GC fraction. Candidate selection is deterministic, using GC-adjusted association followed by validation insertion effects. These runs do not use the LLM.

The selected candidate matches CACGTG in all 12 non-null runs. All six confounded predictors show supported GC effects under the benchmark's 0.05 margin, while shifted AUROC falls to 0.739–0.786 from IID values of 0.993–0.998. This aligns the controlled intervention with a known data shortcut, without assuming that the planted generator fully specifies learned behavior.

The null regime supplies a particularly useful failure case. All three forests have near-chance IID AUROC, yet their selected motif interventions show supported model effects of 0.085–0.128. These responses can reflect fitted spurious structure or out-of-distribution edits. They are not biological discoveries and are not automatically false measurements of predictor dependence. Likewise, one motif-regime CNN recovers the planted candidate but has only a 0.0393 mean insertion contrast, classified as practically small. Candidate recovery, predictive accuracy, and the chosen intervention estimand must therefore be evaluated separately. Full results appear in Appendix C.

D.6. Report Failures and Reproducibility

The completed episode contains three rejected report attempts before an accepted report. The event trace records two rejections for wording that treated inconclusive evidence as a null or absence of reliance, followed by a rejection for numerical prose that violated the rule reserving quantitative values for generated tables. These are examples of checks working on observed failures, not a measured error-detection rate: the rules were developed during this case, and there is no blinded set of independent reports.

The original plan text itself still includes overly narrow language such as "neutral backgrounds" and an informal falsification statement that conflates failure to increase with rejection. The executor's mixed-label population and interval-based decisions determine the actual analysis. This residual mismatch illustrates the limits of field-level report checks. Reproducibility auditing reconstructs all 497 origi-

nal pairwise differences, recomputes their intervals, checks plan and split hashes, and replays sampled model predictions. The expanded analyses retain predictions, intervention pairs, seeds, and source hashes. Successful replay establishes numerical traceability, not biological validity or planner superiority.

E. Paired Evaluation Protocol and Scope

The paired evaluation freezes one task head for each dataset–encoder setting. All four policies and all three repeats share this head and identical sequence identifiers in every partition. The original train/test boundary is preserved. Training, validation, and exploration contain 1,600, 320, and 256 sequences sampled from the original training set. Confirmation contains 256 sequences and audit contains 512 different sequences sampled from the original test set. Selection is stratified with seed 42. All sequences are checked for exact and reverse-complement duplication after preprocessing. Longer sequences are explicitly center-cropped to 500 bases to fit both encoders; this affects 6,880 of 6,914 fruit-fly source sequences and none of the 27,791 human enhancer sequences. The analysis therefore concerns the cropped task, not a full-context benchmark leaderboard.

DNABERT-2 uses base-length-weighted final-token pooling; DNABERT-6 uses mean pooling over overlapping six-mer states, excluding special tokens. Both encoders are frozen and use the same standardized logistic-head selection rule. The original DNABERT checkpoint is pinned at revision `c56e67e`; its deterministic position-ID buffer is constructed by the standard BERT implementation because the older checkpoint omits that non-trained buffer. No learned parameters are left uninitialized.

The structured planner receives the application workflow prompt. The generic planner receives a neutral biomedical-assistant prompt with the same reporting contract, scientific caution, tool schemas, access boundaries, and budgets. The generic prompt omits the prescribed workflow and explicit competing-hypothesis guidance. Thus this comparison isolates a prompt-policy package, not a single component or a different LLM. Both use the same GLM endpoint with temperature 0.2; provider sampling is not controlled by a reproducible API seed. Seeds 7, 17, and 29 control intervention randomness and deterministic-policy choices. The fixed policy chooses the top three exploration-ranked six-mers. The random policy samples three of the same top-20 candidates without replacement. Both include the GC alternative and produce templated reports through the same checks.

All policies have a cap of 8,000 requested prediction sequences, counted before model-cache lookup. Agent calls have a cap of 14 turns and 4,096 generated tokens per turn, with the same per-tool quotas. This matches caps rather than realized resource consumption or total context tokens. Predictor preparation is performed once per cell and excluded from planner costs. Policy order is permuted deterministically within each repeat; shared embedding caches still make wall-clock comparisons approximate. Failed attempts and report retries remain in the resource totals. The recorded usage includes both prompt and completion tokens. The study does not claim equal accelerator consumption across encoders or policies.

Only after every policy episode in a task–encoder cell has ended does the independent auditor open its reserved sequences. Each unique selected motif or GC contrast is replayed with edit seeds 1047 and 2047. Effects are first averaged within source sequence over the two edit seeds; 2,000 bootstrap resamples then operate on source means. The interval level uses the original episode's frozen endpoint-family size. Identical contrasts are computed once and reused across plans, so repeated motifs and episodes are correlated. Audit data are not returned to the planner. Report completion, original support, audit support with the same direction, practical-decision agreement, and resources are retained for every assigned episode.

A replayed effect is evidence about the same predictor under another sampled background population and edit randomness, not an external assay or a newly established biological mechanism. The three datasets are related genomic classification tasks and lack a chromosome/homology grouping audit in this sequence-only evaluation. Both encoders are BERT-family models with different tokenizations. Results therefore do not establish broad architecture or modality transfer. Blinded human report assessment, independent biological validation, and additional model families remain future work.

Retained verification. The final audit verifies all 72 assigned episodes, 32,482 confirmation pairs, and 79,746 audit pairs. It checks frozen protocol, dataset, prompt, split, and plan hashes; disjoint audit membership; identical partitions within each cell; sequence-request and call caps; paired probability differences; motif-edit composition and occurrence constraints; and reconstruction of all audit bootstrap intervals. A further 72 sampled treated/control predictions were reproduced from the six saved task heads and their pretrained encoders within an absolute tolerance of 3×10^{-5}. This sampled numerical replay is not an independent model implementation. The canonical commands are `scripts/prepare_evaluation.py`, `python -m biointerpret.evaluation`, `scripts/summarize_evaluation.py`, and `scripts/verify_evaluation.py -replay`. The release retains all episode traces, failed report at-

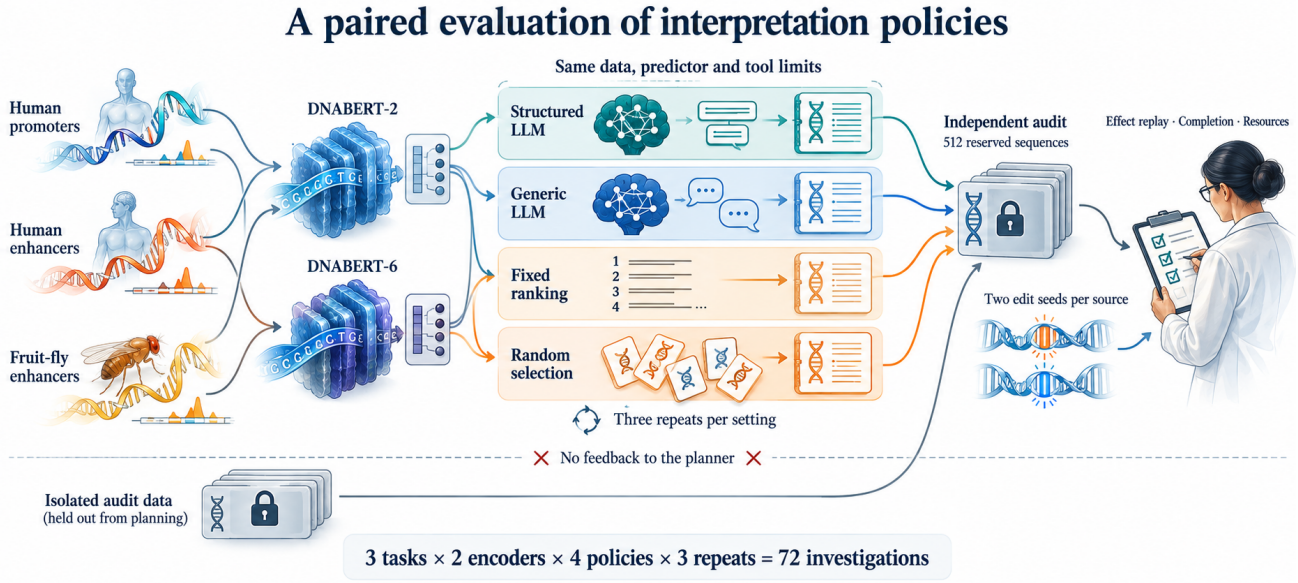


Figure 7. Conceptual design of the original 72-episode paired evaluation, illustrated with GPT Image; the later GenomeOcean extension is not depicted. All three tasks are crossed with both encoders and all four policies. The isolated audit population is opened after planning and does not provide feedback. The inspector is a schematic of evaluation, not a claim that a blinded human study was performed. The four policies share resource caps rather than equal realized costs.

tempts, plans, predictions, raw intervention ledgers, and the frozen protocol.

F. GenomeOcean Extension and Predictor Comparison

The original GenomeOcean-4B base checkpoint is pinned to `DOEJGI/GenomeOcean-4B` revision `2bed2fc`. It is distinct from later continued-training releases. Acquisition verifies the published weight-file hashes and retains hashes for the tokenizer, configuration, and checkpoint shards. The standard Hugging Face Mistral implementation loads every decoder parameter; the unused language-model output head is excluded. No checkpoint-specific remote Python code is executed.

Inference uses an NVIDIA L40S with unquantized bfloat16 weights and native PyTorch scaled dot-product attention. Pooling accumulates in float32 and the logistic head uses the same numerical libraries and selection rule as the primary evaluation. A dedicated worker environment retains PyTorch 2.4, NumPy 1.24, and scikit-learn 1.3, while using Transformers 4.44 and Tokenizers 0.19 to support Mistral. The primary BERT inference uses float32 and Transformers 4.28. These implementation and precision differences are recorded rather than treated as identical accelerator conditions.

The extension configuration is frozen separately in `configs/evaluation_genomeocean.json`, links to the primary protocol hash, and explicitly states that the original outcomes had already been inspected. It retains the three tasks, four policies, three repeats, intervention margins, and reserved audit sequences. It does not retroactively modify the original protocol. All 36 assignments are included regardless of their outcome. The numerical replay tolerance is specified as 0.002 probability before evaluation because mixed-precision inference can vary with batch padding; observed replay discrepancies are retained.

Predictor comparison uses 4,000 stratified bootstrap resamples with seed 2027. Sampled confirmation indices are shared across all three models within each task, yielding paired AUROC differences. Brier score and log loss use the full retained prediction vectors. These post-hoc comparisons are exploratory and unadjusted for multiplicity; they are not a model-selection benchmark or an estimate of model-training uncertainty. The canonical analysis is `scripts/compare_encoders.py` and retains the metrics and paired intervals in `results/evaluation/encoder_comparison.json`.

Extension verification and numerical caveat. The extension audit reconstructs 16,703 confirmation pairs and 37,270 audit pairs. The prespecified changed-batch replay failed its 0.002 probability tolerance on all three tasks: maxima were 0.002011, 0.009379, and 0.007935. A diagnostic follow-up reconstructed original first-uncached batches from the retained audit call order, including repeated-sequence cache reuse. It reproduced all 36 sampled target predictions within 2.3×10^{-16}; this required 256 se-

Policy	Complete	Supported	Replayed	Yield	Tokens	Pred. seq.
Structured LLM	9/9	10	10	1.11	44,213	2,180
Generic LLM	9/9	9	9	1.00	41,460	2,366
Fixed ranking	9/9	9	9	1.00	0	1,460
Random selection	9/9	5	5	0.56	0	1,454

Table 4. GenomeOcean-4B extension (36 investigations). Supported and replayed are endpoint counts, not independent biological discoveries. Yield is replayed supported endpoints per assigned episode, including failures. Tokens and predictor sequences are medians per episode; input and output tokens are included. Deterministic policies use no LLM. All methods share hard access and reporting checks.

Task	Predictor	AUROC [95% interval]	Brier	Log loss
Promoter	DNABERT-2	0.795 [0.741, 0.847]	0.190	0.558
Promoter	DNABERT-6	0.868 [0.822, 0.910]	0.147	0.452
Promoter	GenomeOcean-4B	0.840 [0.792, 0.886]	0.174	0.521
Human enhancer	DNABERT-2	0.791 [0.735, 0.845]	0.193	0.597
Human enhancer	DNABERT-6	0.769 [0.710, 0.826]	0.202	0.598
Human enhancer	GenomeOcean-4B	0.706 [0.643, 0.766]	0.244	0.754
Fly enhancer	DNABERT-2	0.638 [0.568, 0.705]	0.251	0.700
Fly enhancer	DNABERT-6	0.705 [0.640, 0.766]	0.217	0.621
Fly enhancer	GenomeOcean-4B	0.626 [0.558, 0.695]	0.259	0.735

Table 5. Exploratory comparison of the nine fixed predictors on shared confirmation rows. AUROC intervals use 4,000 stratified bootstrap resamples and are unadjusted for multiple comparisons. Brier score and log loss summarize probability error; lower is better. These intervals condition on the fitted heads and omit model-training uncertainty. GenomeOcean was added after the primary study.

quence evaluations across the original batches. This follow-up does not reverse the failed batch-sensitivity diagnostic. It distinguishes reproducibility under recorded batching from invariance to batching. The result is retained as `passed_with_batch_sensitivity`, with separate per-task outcomes. It is not a claim that the mixed-precision implementation is a batch-invariant numerical oracle.

F.1. Fixed-head full-precision sensitivity analysis

After observing the failed bfloat16 rebatching diagnostic, we froze a separate numerical follow-up in `configs/genomeocean_precision.json`. We cast the same checkpoint parameters to float32, disabled TF32 matrix multiplication, and retained native SDPA, the original base-weighted pooling, all three fitted task heads, and every audit edit. Thus this is a post-hoc sensitivity analysis of fixed predictors, not a repeat of model fitting, agent planning, or confirmation. First, we compared the same 12 target predictions per task under regrouped and reconstructed original batches, with a 3×10^{-5} tolerance. Second, we recomputed all 37,270 audit pairs, averaged the same two edit seeds per source, and repeated the original 2,000 bootstrap resamples and endpoint multiplicity adjustment. Repeated endpoints remain correlated. The canonical execution and analysis scripts are `check_genomeocean_precision.py` and `summarize_genomeocean_precision.py`; separate ledgers preserve new probabilities without modifying the original evidence. All three FP32 rebatching checks passed: maximum probability differences were 6.55×10^{-7}, 5.86×10^{-7}, and 1.53×10^{-6} for promoters, human enhancers, and fly enhancers. Across all 37,270 audit pairs, none of the 144 endpoint decisions changed. The 55 audit-supported endpoints remained supported, including the 33 supported at initial confirmation; these counts include repeated contrasts. Maximum endpoint mean-effect changes were 0.00277, 0.00084, and 0.00041, respectively. Individual probabilities differed more substantially between BF16 and FP32, with maxima of 0.1113, 0.0682, and 0.0602 (Table 6). Stable aggregate decisions therefore do not imply invariant individual predictions. This follow-up supports the audit conclusions under a change of inference precision, conditional on the original BF16-trained heads and fixed edits. It neither reverses the original failed BF16 batch check nor establishes batch invariance for untested sequences.

Table 6. Post-hoc FP32 audit with fixed heads and edits. Δp compares all recomputed audit probabilities against stored BF16 values; $\Delta\bar{d}$ compares endpoint mean effects. Verdict counts include repeated, correlated endpoints.

Task	Max $\lvert\Delta p\rvert$	Max $\lvert\Delta\bar{d}\rvert$	Changed
Promoter	0.1113	0.0028	0/48
Human enhancer	0.0682	0.0008	0/48
Fly enhancer	0.0602	0.0004	0/48

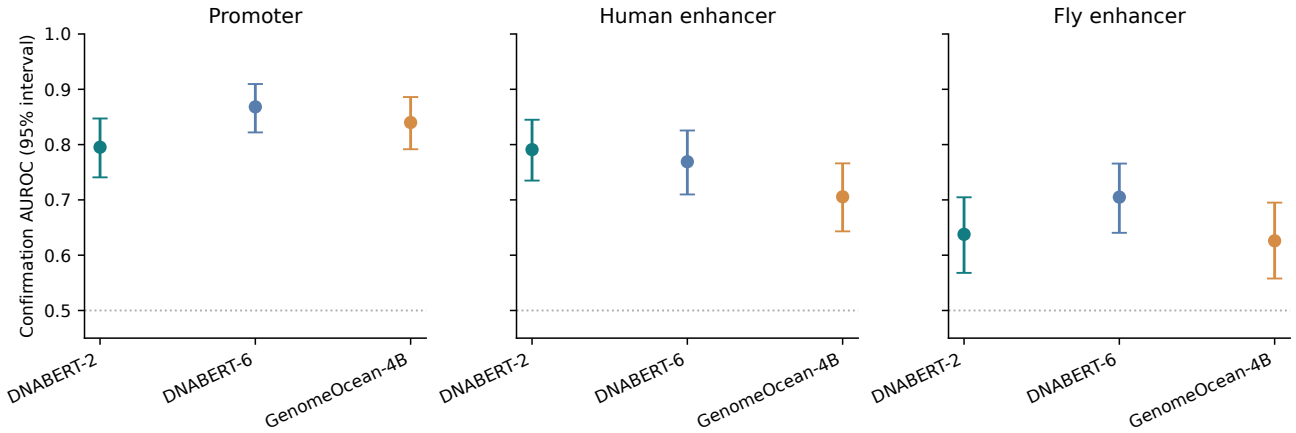


Figure 8. Exploratory comparison of all three genomic backbones with fitted task heads. Each task uses the same 256 confirmation sequence identities across models. Error bars are unadjusted 95% stratified bootstrap intervals, conditional on each fitted predictor; they do not measure independent training variation or biological interpretation accuracy. GenomeOcean was added after inspecting the original two-model results.

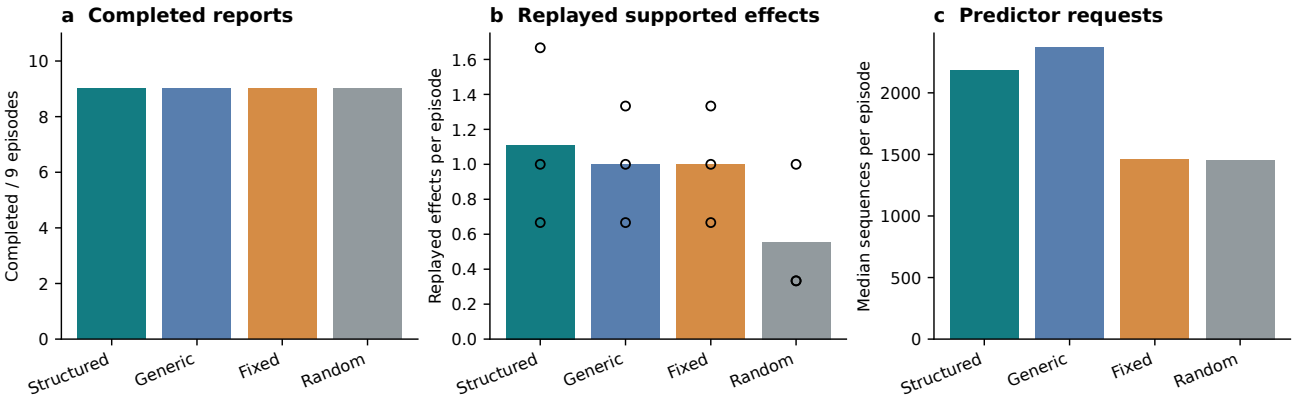


Figure 9. GenomeOcean-4B extension: nine investigations per policy over three tasks and three repeats. Open circles are task means, not uncertainty intervals. The shared caps do not imply equal realized costs. These results are retained separately from the original 72-episode aggregate.

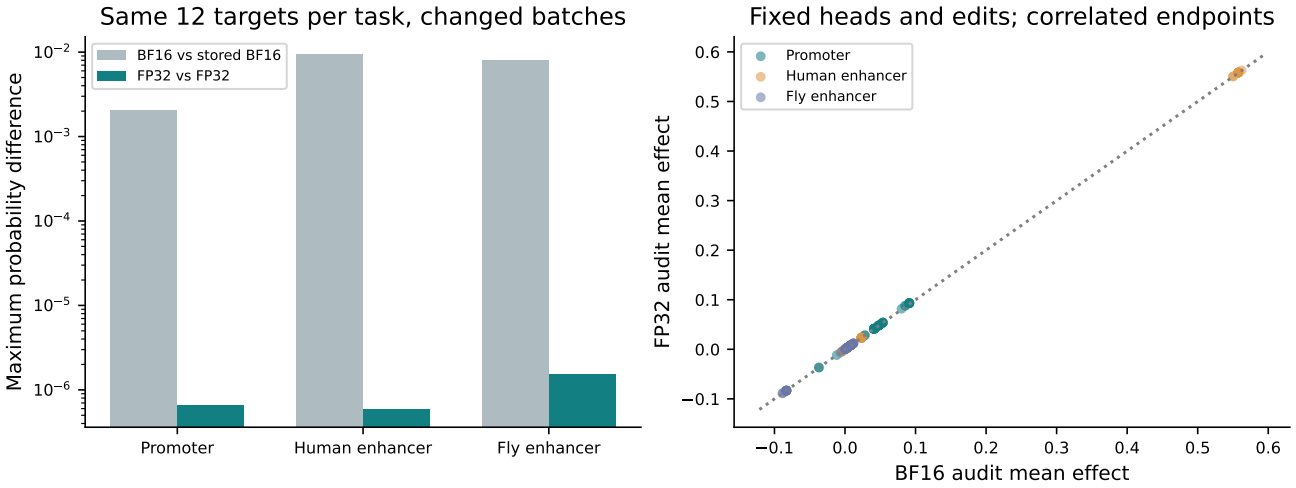


Figure 10. Measured numerical sensitivity of GenomeOcean-4B. Left: changed-batch probability differences for the same 12 target predictions per task, comparing each precision against its own original-batch calculation. Right: all 144 audit mean effects under the fixed-head precision change. Points overlap because policies can select identical contrasts. The diagonal denotes equal effects; endpoint uncertainty and decisions use the retained source-level bootstrap.